

Penalty view

prices weight costs

$$\lambda_k = \beta_k$$

graph price off ($c = 0$)

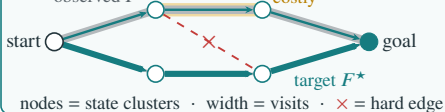
β fixed $\rightarrow$ fixed penalty

β adaptive $\rightarrow$ Lagrangian

exact settings of the
same price rule

SafeOT-Dual: one price rule

① each epoch: route visits to reward, within budgets
observed $\hat{F}$



② price

$$\lambda_k = \bar{\lambda}_k^* + \beta_k$$

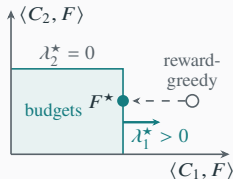
solved, clipped at c , filtered

feedback on $\bar{J}_k - B_k$

③ actor $\tilde{A} = \frac{A_r - \sum_k \lambda_k A_{c_k}}{1 + \sum_k \lambda_k} \rightarrow$ PPO/TRPO

Constraint view

prices are budget duals



$\varepsilon \rightarrow 0$: LP dual price
saturated clip: active-set price
limits, under Sec. 4 assumptions

Payoff while learning. Cost/budget during training (task median); shaded = violation V .

— SafeOT-Dual ··· TRPO-Lag

